

For this project I built a small weather analytics pipeline for a fictional regional operations team. I used the Open-Meteo Historical Weather API to pull daily weather data for three cities over a full year, focusing on temperature, precipitation, and wind speed.

On the ingestion side, I wrote a Python script with the requests library to call the archive endpoint for each city and save the raw JSON responses into a data folder. Those files then feed into a flatten-and-clean step where I load the JSON with pandas, turn the nested daily arrays into a row-based DataFrame, drop any null values, and remove duplicate city/date combinations before exporting a clean CSV.

For storage and analysis, I designed a normalized PostgreSQL schema with a `weather.cities` table and a `weather.weather_records` table. Cities store basic location information, and `weather_records` store the daily measurements with a foreign key back to the city. I used `psycopg2` and simple DAO classes to create the schema, insert cities, and load all of the cleaned daily records into the database.

Once the data was loaded, I wrote SQL queries to answer business-style questions about the year of weather data. These include the largest and average daily temperature gaps per city, the wettest single day per city, how many days per city exceeded 100°F, and how many days were rainy versus completely dry. Together, these queries show which cities are more volatile in temperature, which are hotter or wetter overall, and how often each city experiences extreme or challenging conditions.

Throughout the project I kept everything in a GitHub repository with clear commits for each stage: API ingestion, JSON flattening and cleaning, schema creation, database loading, and SQL analysis. This makes the project easy to review end-to-end and shows the full flow from raw external data to cleaned, queryable tables and concrete insights.